

# Kanay Gupta

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## EDUCATION

**University of New South Wales (UNSW)**

February 2023 – Expected December 2026

Bachelor of Advanced Computer Science (Honours)

- WAM: 87 (High Distinction). Faculty of Engineering Dean's Honours List (2023).

## EXPERIENCE

**ResMed**

May 2026 – Present

*Student Intern – Software Engineer, Digital Platform Engineering*

*Tech Stack: Java, TypeScript/Angular, Python, Terraform, AWS (EKS, ECS), Docker, GitHub Actions*

- Delivered 20+ features and defect fixes across four services of a cloud-hosted device simulation platform used company-wide for device verification and validation, spanning a Java REST API, an Angular portal, and Python services.
- Built end-to-end simulation support for a new generation of connected medical devices, adding settings acknowledgement, schema-version-aware IoT endpoint routing, filter presets, and a review-and-retry flow letting users edit and resend failed device signals.
- Resolved defects across the Java API layer, correcting spurious 400 responses on optional parameters, adding global exception handling for failed type conversions, and reconciling mismatched product names generated for one device by two services.
- Migrated the web portal and public API onto the company's shared Kubernetes (EKS) deployment platform, reworking GitHub Actions pipelines and validating routing across development, staging, and production.

**Satire Explanation with Language Models (Honours Thesis)**

February 2026 – Present

*Undergraduate Honours Researcher (UNSW Computer Science and Engineering)*

*Tech Stack: Python, PyTorch, HuggingFace, OpenRouter, Qwen2-VL, Llama-3.1, Mistral, Slurm/HPC*

- Curated a 1,074-article dataset of Australian satirical and factual news to test whether Large Language Models (LLMs) can *explain* satire rather than merely classify it, addressing prior work limited to US-centric cartoon corpora.
- Designed an “unreliable narrator” prompting framework grounded in narrative theory that raised judged explanation quality by 0.26–0.80 points on a 5-point scale across all four model families tested.
- Built an LLM-as-judge evaluation harness with Prometheus-style rubrics scoring 32 generation runs across two datasets, four models, and four context conditions, validated by a blind dual-annotator study.
- Established two negative results by ablation: a sentence-level pre-classifier lowered event-grounding scores in seven of eight conditions, and article images improved grounding without improving explanation quality.
- Preparing a paper for ALTA 2026 archival submission covering the generalisation finding, null results, and failure modes, supervised by Dr. Aditya Joshi.

## PROJECTS

**LedgerMind – AI-Powered Tax Compliance App**

September 2025 – November 2025

*Project Manager and Engineer (Capstone Project for IntelliServe)*

*Tech Stack: Python, React Native, PostgreSQL, Docker, OpenAI API, RAG, Jira, Git*

- Led a 6-person agile team through weekly sprints and stakeholder meetings to deliver a functional MVP for an AI-driven tax deduction mobile app.
- Built an LLM extraction pipeline pulling vendor, date, and amount details from receipt images, voice logs, and email invoices, removing manual data entry.
- Designed a Retrieval-Augmented Generation (RAG) system validating expenses against live Australian Taxation Office (ATO) feeds, with compliance-ready PDF audit exports.

**AI-Powered Job Ad Extractor – LLM Comparative Analysis**

March 2025 – April 2025

*AI Engineer (Industry Project for SEEK)*

*Tech Stack: Python, PyTorch, HuggingFace, Scikit-learn, Pandas, Gemini API*

- Benchmarked three approaches to numerical extraction from job ads, then fine-tuned an open-weight Pythia-160m model with custom prediction heads, cutting Mean Absolute Error by over 95% against a Scikit-learn baseline.
- Annotated a foundational dataset of 50+ samples, achieving 94% inter-annotator agreement to establish a reliable evaluation benchmark.

**Ray Tracing Engine from Scratch in Rust – [GitHub](#)**

July 2025

*Tech Stack: Rust, Multithreading*

- Wrote a multithreaded path tracing engine from scratch without external rendering crates, implementing physically based materials (metals, dielectrics), texture mapping, and a configurable depth-of-field camera.
- Achieved a 7.16x rendering speedup on a 16-core CPU by parallelising the workload using Rust's native concurrency primitives (Arc, Mutex).

## SKILLS

- **Languages:** Python, Java, TypeScript/JavaScript, Rust, C#, SQL, HTML/CSS
- **Frameworks and Libraries:** PyTorch, HuggingFace, Scikit-learn, Pandas, Angular, React.js, Next.js, Node.js, Tailwind CSS
- **Tools and Platforms:** AWS (EKS, ECS), Terraform, Docker, GitHub Actions, Git, PostgreSQL, Linux, Bash, Jira, Slurm, LaTeX